

AKHIL YADAV

Controls Engineer | SCADA, RTU and Controls Systems - Renewable Energy and Data Centers

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EU Work Authorisation: Available | Open to relocation | Available Immediately

PROFESSIONAL SUMMARY

Controls engineer with 5+ years of SCADA, RTU, PLC and DCS work on offshore oil and gas, offshore wind and solar EPC projects. Raised DCS reliability (MTBF) 43% by leading a 45+ module ABB 800xA software migration; held 99.7% uptime on STARDOM RTU to CENTUM DCS telemetry (Modbus, DNP3, OPC) for a USD 250M Qatar Energy programme and closed 38 quality exceptions through corrective-action plans. Reviewed inverter, transformer-station and SCADA controls for 350 MWp across 6 solar sites. MBA, TU Berlin.

TECHNICAL SKILLS

Controls & SCADA: SCADA, PLC and RTU systems: ABB 800xA, Yokogawa CENTUM VP/C3000, STARDOM FCN RTUs, FAST/TOOLS SCADA; IEC 61131-3 (Structured Text, FBD, Ladder), function-block libraries, HMI graphics

Protocols & Networks: IEC 61850, DNP3, Modbus TCP/RTU, OPC UA/DA, TCP/IP, HART; redundant fibre and radio telemetry links

High-Voltage & Plant Equipment: Inverters, transformer stations, controllers, I/O systems, telecommunications links, SCADA packages from OEMs (Huawei, SMA, KACO, Gantner)

Root Cause & Reliability: Root cause analysis (RCA), corrective and preventive actions, trip, communication-loss and redundancy-loss diagnosis (LOV/LOR-type events), server redundancy (HAC), MTBF and availability tracking, alarm rationalisation

Software Lifecycle & Documentation: Software migration, FAT/SAT, MODSIM testing, loop checks, punch-list closure, control narratives and functional descriptions, I/O lists, ISA datasheets, cable schedules, SOPs

Functional Safety: IEC 61511 lifecycle, IEC 61508, SIL-rated shutdown interlocks, ESD, fire and gas cause and effect, HAZOP/SIL support, NAMUR NE 107

Tools & Certifications: Python, Power BI, MS Project, Excel, AutoCAD, EPLAN (familiar), SAP Ariba; IEC 61511 Functional Safety and ABB 800xA Applied Operations certified

PROFESSIONAL EXPERIENCE

Controls Engineer, EPC Solar | ib vogt GmbH | Berlin

Apr 2025 - Sep 2025

350 MWp solar EPC portfolio, 6 concurrent sites; GBP 9M+ of inverter, transformer-station, SCADA and balance-of-plant packages.

Inverter, Transformer-Station & SCADA Package Review

- Benchmarked inverter and transformer-station bids against 66 projects (3.4 GWp, 15 countries; mean 4.6 cent/Wp) by sizing each package on DC/AC ratio (median 1.16) and capacity.
- Narrowed 4 SCADA bidders (38-76% price spread) to a clear award case by pricing every bid at four project sizes (10-150 MWp).

Customer, Subcontractor & OEM Technical Support

- Raised on-time delivery from 76% to 97% across 6 sites by tracking OEM submittals and closing technical queries (RFIs) with Huawei, SMA, KACO and Gantner.
- Secured GBP 9M+ across 6 OEM contracts with 66-month inverter warranties, 5% warranty bonds and late-delivery penalties of 0.2-1% per day capped at 5-15%; cut the supplier base from 9 to 6.

Controls Design Review, Documentation & Hand-off

- Cut NCRs 18% and avoided a 2-week commissioning delay by reviewing schematics and I/O lists for 42 control panels (680+ I/O points) before construction.
- Handed over 6 sites on agreed milestones via weekly cost, schedule and quality KPIs (Power BI, SAP Ariba), with zero compliance failures.

Product Engineer, Process Automation | ABB | Bengaluru

Jan 2023 - Sep 2024

Lead on 3 concurrent DCS and automation projects, USD 12M combined: software migration, logic conversion, testing and vendor coordination.

Software Migration & Testing

- Raised MTBF 43% by leading the migration of 45+ legacy ProControl P14 modules to ABB 800xA, converting C-based logic to IEC 61131-3 Structured Text built as standard, reusable function blocks.
- Closed 38 control-level defects across 1,200+ I/O signals with zero critical defects at go-live by running FAT/SAT from audit-ready I/O lists and ISA sheets.

Availability, Alarms & Redundancy

- Held DCS uptime above 99.9% on 3 live projects by configuring server redundancy (HAC), the historian and OPC DA/AE/UA links, and alarm rationalisation to NAMUR NE 107.
- Cut HMI build effort ~25% across six process stages by setting standard naming, layout and faceplate conventions for 800xA graphics built to the design data sheets.

Vendor Coordination & Documentation

- Held vendor on-time delivery above 94% (Siemens, Schneider Electric, Emerson), clearing every SLA breach within 48 hours, by directing multi-discipline controls, SCADA and OEM teams; wrote the SOPs and training adopted by operations.

Project Engineer, EPC Automation | Yokogawa | Bengaluru

Sep 2020 - Dec 2022

USD 250M Qatar Energy offshore EPC: telemetry and safety logic for the Maydan Mahzam and Bul Hanine fields, 40+ wellhead jackets.

RTU & Telemetry Integration

- Achieved 99.7% uptime and critical alarm response under 50 ms by linking STARDOM RTUs to the CENTUM DCS at two production stations over OPC, with redundant fibre and radio links.
- Removed IP conflicts and tag-mapping errors before FAT by pre-testing Modbus RTU, DNP3 and OPC links with MODSIM; added wireless telemetry for 3 unpowered jackets.

Safety Logic, Root Cause & Corrective Actions

- Reached zero open safety items across 8 SATs by verifying fire and gas and process cause and effect matrices against I/O schedules for every jacket, to the IEC 61511 lifecycle.
- Engineered the 16-step remote well-opening sequence for 4 jackets (13 wells) in STARDOM RTUs with SIL-rated shutdown interlocks and timed logic (48-hour standby auto-trip).
- Closed 38 quality exceptions through corrective-action plans while holding 8 OEM vendors (Emerson, Honeywell, ABB, Schneider Electric) to 91%+ on-time delivery across India and Qatar.

Testing & Documentation

- Cut defect density 35% and reached a 93% first-pass rate (~1,700 of 1,850+ loop checks right first time) across 14 FATs and 8 SATs by standardising test procedures, with zero delays at final acceptance.
- Delivered 180+ ISA datasheets, 65 panel BOMs and 3,200+ cable schedules with zero rework; control narrative approved at Rev 4 (Code 10).

EDUCATION

MBA, Energy Management | TU Berlin, Germany

- Thesis: green hydrogen cost model across 9 countries (8,760 hourly steps each); a 1% rise in financing cost cancels a 10% gain in renewable output.

B.Tech, Electronics and Instrumentation Engineering | Amrita Vishwa Vidyapeetham, India

- Publication: "ML Algorithms for Gas Leak Detection", Springer Singapore, 2020.

LANGUAGES & AVAILABILITY

English: C2 (business proficient) | German: B1 (intermediate) | EU work authorisation, available immediately.